

2.0.5-hpi-lebesgue-dev

September 2, 2024

```
[15]: %load_ext autoreload
      %autoreload 2

      import cProfile
      import pstats
      import io
      import os
      from datetime import datetime
      import tracemalloc
      import torch
      import pandas as pd
      import psutil
      from memory_profiler import profile
      import time

      from prism.config import PROCESSED_DATA_DIR, MODELS_DIR
      from prism.PRISM_functions import normalise
      from prism.partial_responses import PartialResponseCalculator
      from prism.load_models import load_mlp
```

The autoreload extension is already loaded. To reload it, use:

```
%reload_ext autoreload
```

```
[16]: # Load and preprocess data
      data_train = pd.read_csv(PROCESSED_DATA_DIR.joinpath('imputed_dataset1_train.
      ↪csv'))
      data_test = pd.read_csv(PROCESSED_DATA_DIR.joinpath('imputed_dataset1_test.
      ↪csv'))

      # drop id column and target column
      target_col = 'oneyearmort'
      data_train.drop(['trr_id_code', target_col], axis=1, inplace=True)
      data_test.drop(['trr_id_code', target_col], axis=1, inplace=True)

      # Normalize data
      x_train, x_test = normalise(data_train, data_test)

      # Load MLP model
```

```

filename_mlp = 'mlp_model_20240705_135534'
try:
    mlp, mlp_params, _ = load_mlp(filename_mlp, MODELS_DIR)
    print("Model loaded successfully")
except Exception as e:
    print(f"Error loading model: {e}")
    raise

# Extract method and device from mlp_params, with defaults
method = mlp_params.get('method', 'dirac')
device = mlp_params.get('device', 'cpu')

```

Model loaded successfully

```

[19]: @profile
def _calculate_lebesgue_profiled(calculator, x):
    return calculator._calculate_lebesgue(x)

def run_profiling(n_samples=500):
    # Use the first n_samples from x_train
    x = torch.tensor(x_train.iloc[:n_samples].values, dtype=torch.float32).
    ↪to(device)

    # Initialize PartialResponseCalculator with the loaded MLP model
    calculator = PartialResponseCalculator(mlp, method='lebesgue',
    ↪device=device, input_dim=x.shape[1], x_train=x)

    # Start memory tracing
    tracemalloc.start()

    # Profile the _calculate_lebesgue method
    pr = cProfile.Profile()
    pr.enable()

    # Run the method
    _calculate_lebesgue_profiled(calculator, x)

    pr.disable()

    # Get memory usage
    current, peak = tracemalloc.get_traced_memory()
    tracemalloc.stop()

    # Get CPU usage
    cpu_percent = psutil.cpu_percent()

    # Get GPU usage if available

```

```

gpu_percent = None
if device == 'cuda' and torch.cuda.is_available():
    gpu_percent = torch.cuda.utilization()

# Create a string buffer to capture the output
s = io.StringIO()
ps = pstats.Stats(pr, stream=s).sort_stats('cumulative')
ps.print_stats()

# Get the profiling results as a string
profiling_results = s.getvalue()

# Add memory and CPU/GPU usage to the results
profiling_results += f"\nMemory usage: current={current/1e6:.1f}MB, \u2192
peak={peak/1e6:.1f}MB\n"
profiling_results += f"CPU usage: {cpu_percent}%\n"
if gpu_percent is not None:
    profiling_results += f"GPU usage: {gpu_percent}%\n"

# Print the results to console
print(profiling_results)

# Save the results to a file and return the filepath
return save_results(profiling_results, n_samples, device)

def save_results(results, n_samples, device):
    # Create a 'profiling_results' directory within MODELS_DIR if it doesn't \u2192
    exist
    profiling_dir = os.path.join(MODELS_DIR, 'profiling_results')
    if not os.path.exists(profiling_dir):
        os.makedirs(profiling_dir)

    # Generate a filename with timestamp, sample size, and device
    timestamp = datetime.now().strftime("%Y%m%d_%H%M%S")
    filename = f"lebesgue_profile_{timestamp}_samples{n_samples}_{device}.txt"
    filepath = os.path.join(profiling_dir, filename)

    # Save the results to the file
    with open(filepath, 'w') as f:
        f.write(results)

    print(f"Profiling results saved to: {filepath}")
    return filepath

```

```

[25]: # torch.no_grad() in _calculate_lebesgue
%reload_ext autoreload
filepath = run_profiling(n_samples=500)

```

ERROR: Could not find file C:\Users\Henry
Pigot\AppData\Local\Temp\ipykernel_93128\385629262.py
3714324 function calls (3356681 primitive calls) in 264.267 seconds

Ordered by: cumulative time

ncalls	tottime	percall	cumtime	percall	filename:lineno(function)
1	0.000	0.000	264.267	264.267	c:\Users\Henry Pigot\Documents\MEGAsync\2024 post doc\explainable survival prediction model\prism_github\venv_prism\Lib\site-packages\memory_profiler.py:1185(wrapper)
1	0.000	0.000	264.265	264.265	c:\Users\Henry Pigot\Documents\MEGAsync\2024 post doc\explainable survival prediction model\prism_github\venv_prism\Lib\site-packages\memory_profiler.py:759(f)
1	0.011	0.011	264.265	264.265	C:\Users\Henry Pigot\AppData\Local \Temp\ipykernel_93128\385629262.py:1(_calculate_lebesgue_profiled)
1	27.409	27.409	264.254	264.254	C:\Users\Henry Pigot\Documents\MEGAsync\2024 post doc\explainable survival prediction model\prism_github\prism\partial_responses.py:100(_calculate_lebesgue)
27511	5.866	0.000	224.073	0.008	C:\Users\Henry Pigot\Documents\MEGAsync\2024 post doc\explainable survival prediction model\prism_github\prism\partial_responses.py:32(predict)
27511	8.626	0.000	200.145	0.007	C:\Users\Henry Pigot\Documents\MEGAsync\2024 post doc\explainable survival prediction model\prism_github\prism\maskedmlp.py:42(predict)
110044/27511	7.436	0.000	108.120	0.004	c:\Users\Henry Pigot\Documents\MEGAsync\2024 post doc\explainable survival prediction model\prism_github\venv_prism\Lib\site- packages\torch\nn\modules\module.py:1528(_wrapped_call_impl)
110044/27511	20.162	0.000	106.115	0.004	c:\Users\Henry Pigot\Documents\MEGAsync\2024 post doc\explainable survival prediction model\prism_github\venv_prism\Lib\site- packages\torch\nn\modules\module.py:1534(_call_impl)
27511	8.978	0.000	100.459	0.004	C:\Users\Henry Pigot\Documents\MEGAsync\2024 post doc\explainable survival prediction model\prism_github\prism\maskedmlp.py:36(forward)
27511	1.381	0.000	70.909	0.003	c:\Users\Henry Pigot\Documents\MEGAsync\2024 post doc\explainable survival prediction model\prism_github\venv_prism\Lib\site- packages\torch\nn\modules\module.py:2433(eval)
110044/27511	17.149	0.000	69.528	0.003	c:\Users\Henry Pigot\Documents\MEGAsync\2024 post doc\explainable survival prediction model\prism_github\venv_prism\Lib\site- packages\torch\nn\modules\module.py:2411(train)
27511	1.081	0.000	32.818	0.001	c:\Users\Henry Pigot\Documents\MEGAsync\2024 post doc\explainable survival prediction model\prism_github\venv_prism\Lib\site- packages\torch\nn\modules\activation.py:356(forward)
27511	31.736	0.001	31.736	0.001	{built-in method torch.tanh}

```

110044 19.478 0.000 30.221 0.000 c:\Users\Henry
Pigot\Documents\MEGAsync\2024 post doc\explainable survival prediction
model\prism_github\venv_prism\Lib\site-
packages\torch\nn\modules\module.py:1711(__setattr__)
55022 6.647 0.000 25.076 0.000 c:\Users\Henry
Pigot\Documents\MEGAsync\2024 post doc\explainable survival prediction
model\prism_github\venv_prism\Lib\site-
packages\torch\nn\modules\linear.py:115(forward)
192577 9.610 0.000 22.015 0.000 c:\Users\Henry
Pigot\Documents\MEGAsync\2024 post doc\explainable survival prediction
model\prism_github\venv_prism\Lib\site-
packages\torch\nn\modules\module.py:2313(children)
220088 20.974 0.000 20.974 0.000 c:\Users\Henry
Pigot\Documents\MEGAsync\2024 post doc\explainable survival prediction
model\prism_github\venv_prism\Lib\site-
packages\torch\nn\modules\module.py:1696(__getattr__)
192577 11.868 0.000 12.405 0.000 c:\Users\Henry
Pigot\Documents\MEGAsync\2024 post doc\explainable survival prediction
model\prism_github\venv_prism\Lib\site-
packages\torch\nn\modules\module.py:2322(named_children)
55022 12.198 0.000 12.198 0.000 {built-in method
torch._C._nn.linear}
495203/385159 3.665 0.000 10.542 0.000 {built-in method
builtins.isinstance}
110044 6.713 0.000 7.527 0.000 c:\Users\Henry
Pigot\Documents\MEGAsync\2024 post doc\explainable survival prediction
model\prism_github\venv_prism\Lib\site-
packages\torch\nn\parameter.py:8(__instancecheck__)
55024 3.593 0.000 7.042 0.000 c:\Users\Henry
Pigot\Documents\MEGAsync\2024 post doc\explainable survival prediction
model\prism_github\venv_prism\Lib\site-
packages\torch\autograd\grad_mode.py:79(__enter__)
27511 1.973 0.000 6.716 0.000 c:\Users\Henry
Pigot\Documents\MEGAsync\2024 post doc\explainable survival prediction
model\prism_github\venv_prism\Lib\site-packages\torch\_tensor.py:34(wrapped)
110048 5.537 0.000 6.566 0.000 c:\Users\Henry
Pigot\Documents\MEGAsync\2024 post doc\explainable survival prediction
model\prism_github\venv_prism\Lib\site-
packages\torch\autograd\grad_mode.py:183(__init__)
27511 1.792 0.000 6.315 0.000 {built-in method builtins.hasattr}
55024 2.941 0.000 6.224 0.000 c:\Users\Henry
Pigot\Documents\MEGAsync\2024 post doc\explainable survival prediction
model\prism_github\venv_prism\Lib\site-
packages\torch\autograd\grad_mode.py:83(__exit__)
55024 4.505 0.000 5.685 0.000 c:\Users\Henry
Pigot\Documents\MEGAsync\2024 post doc\explainable survival prediction
model\prism_github\venv_prism\Lib\site-
packages\torch\autograd\grad_mode.py:74(__init__)

```

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27511    0.922    0.000    4.672    0.000 c:\Users\Henry
Pigot\Documents\MEGAsync\2024 post doc\explainable survival prediction
model\prism_github\venv_prism\Lib\site-packages\torch\_tensor.py:964(__rsub__)
27511    3.751    0.000    3.751    0.000 {built-in method torch.rsub}
27511    3.248    0.000    3.248    0.000 {method 'mean' of
'torch._C.TensorBase' objects}
55024    2.329    0.000    2.776    0.000 c:\Users\Henry
Pigot\Documents\MEGAsync\2024 post doc\explainable survival prediction
model\prism_github\venv_prism\Lib\site-
packages\torch\utils\_contextlib.py:149(__new__)
27511    2.139    0.000    2.139    0.000 {built-in method torch.sigmoid}
27500    1.469    0.000    1.469    0.000 {method 'clone' of
'torch._C.TensorBase' objects}
55023    1.378    0.000    1.378    0.000 {method 'to' of
'torch._C.TensorBase' objects}
110044   1.290    0.000    1.290    0.000 {built-in method
torch._C._get_tracing_state}
27511    1.245    0.000    1.245    0.000 {built-in method torch.log}
55024    1.179    0.000    1.179    0.000 c:\Users\Henry
Pigot\Documents\MEGAsync\2024 post doc\explainable survival prediction
model\prism_github\venv_prism\Lib\site-
packages\torch\_jit_internal.py:1120(is_scripting)
27511    1.051    0.000    1.051    0.000 {method 'squeeze' of
'torch._C.TensorBase' objects}
110048   0.849    0.000    0.849    0.000 {built-in method
torch._C._set_grad_enabled}
55024    0.447    0.000    0.447    0.000 {built-in method __new__ of type
object at 0x00007FF804FA8F90}
110044   0.435    0.000    0.435    0.000 {method 'items' of
'collections.OrderedDict' objects}
330136   0.428    0.000    0.428    0.000 {method 'get' of 'dict' objects}
165072   0.347    0.000    0.347    0.000 {built-in method
torch.is_grad_enabled}
110044   0.163    0.000    0.163    0.000 {function
_ParameterMeta.__instancecheck__ at 0x00000246EEFA0720}
82533    0.102    0.000    0.102    0.000 {method 'add' of 'set' objects}
22       0.083    0.004    0.083    0.004 {method 'repeat' of
'torch._C.TensorBase' objects}
27511    0.071    0.000    0.071    0.000 {built-in method
torch._C._has_torch_function}
56       0.004    0.000    0.004    0.000 {built-in method torch.zeros}
1       0.001    0.001    0.001    0.001 {built-in method torch.stack}
22       0.001    0.000    0.001    0.000 {method 'reshape' of
'torch._C.TensorBase' objects}
1       0.000    0.000    0.001    0.001 c:\Users\Henry
Pigot\Documents\MEGAsync\2024 post doc\explainable survival prediction
model\prism_github\venv_prism\Lib\site-packages\memory_profiler.py:713(__call__)
1       0.000    0.000    0.001    0.001 c:\Users\Henry

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Pigot\Documents\MEGAsync\2024 post doc\explainable survival prediction
model\prism_github\venv_prism\Lib\site-
packages\memory_profiler.py:728(add_function)
    1    0.000    0.000    0.001    0.001 c:\Users\Henry
Pigot\Documents\MEGAsync\2024 post doc\explainable survival prediction
model\prism_github\venv_prism\Lib\site-packages\memory_profiler.py:645(add)
    22    0.001    0.000    0.001    0.000 {method 'unsqueeze' of
'torch._C.TensorBase' objects}
    1    0.000    0.000    0.000    0.000 {built-in method builtins.print}
    2    0.000    0.000    0.000    0.000 c:\Users\Henry
Pigot\Documents\MEGAsync\2024 post doc\explainable survival prediction
model\prism_github\venv_prism\Lib\site-packages\ipykernel\iostream.py:655(write)
    1    0.000    0.000    0.000    0.000 <frozen genericpath>:16(exists)
    1    0.000    0.000    0.000    0.000 {built-in method nt.stat}
   110    0.000    0.000    0.000    0.000 {method 'append' of 'list'
objects}
    2    0.000    0.000    0.000    0.000 {built-in method builtins.next}
    1    0.000    0.000    0.000    0.000 C:\Users\Henry
Pigot\AppData\Local\Programs\Python\Python311\Lib\contextlib.py:141(__exit__)
    2    0.000    0.000    0.000    0.000 c:\Users\Henry
Pigot\Documents\MEGAsync\2024 post doc\explainable survival prediction
model\prism_github\venv_prism\Lib\site-
packages\ipykernel\iostream.py:577(_schedule_flush)
    2    0.000    0.000    0.000    0.000 c:\Users\Henry
Pigot\Documents\MEGAsync\2024 post doc\explainable survival prediction
model\prism_github\venv_prism\Lib\site-
packages\memory_profiler.py:740(_count_ctxmgr)
    1    0.000    0.000    0.000    0.000 c:\Users\Henry
Pigot\Documents\MEGAsync\2024 post doc\explainable survival prediction
model\prism_github\venv_prism\Lib\site-
packages\ipykernel\iostream.py:259(schedule)
    1    0.000    0.000    0.000    0.000 c:\Users\Henry
Pigot\Documents\MEGAsync\2024 post doc\explainable survival prediction
model\prism_github\venv_prism\Lib\site-packages\memory_profiler.py:702(__init__)
    1    0.000    0.000    0.000    0.000 c:\Users\Henry
Pigot\Documents\MEGAsync\2024 post doc\explainable survival prediction
model\prism_github\venv_prism\Lib\site-
packages\memory_profiler.py:748(wrap_function)
    1    0.000    0.000    0.000    0.000 C:\Users\Henry
Pigot\AppData\Local\Programs\Python\Python311\Lib\contextlib.py:132(__enter__)
    1    0.000    0.000    0.000    0.000 c:\Users\Henry
Pigot\Documents\MEGAsync\2024 post doc\explainable survival prediction
model\prism_github\venv_prism\Lib\site-packages\zmq\sugar\socket.py:626(send)
    1    0.000    0.000    0.000    0.000 c:\Users\Henry
Pigot\Documents\MEGAsync\2024 post doc\explainable survival prediction
model\prism_github\venv_prism\Lib\site-
packages\memory_profiler.py:782(disable_by_count)
    1    0.000    0.000    0.000    0.000 c:\Users\Henry

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```

Pigot\Documents\MEGAsync\2024 post doc\explainable survival prediction
model\prism_github\venv_prism\Lib\site-
packages\memory_profiler.py:1201(choose_backend)
    1    0.000    0.000    0.000    0.000 C:\Users\Henry Pigot\AppData\Local
\Programs\Python\Python311\Lib\asyncio\coroutines.py:21(iscoroutinefunction)
    1    0.000    0.000    0.000    0.000 C:\Users\Henry Pigot\AppData\Local
\Programs\Python\Python311\Lib\inspect.py:409(iscoroutinefunction)
    1    0.000    0.000    0.000    0.000 c:\Users\Henry
Pigot\Documents\MEGAsync\2024 post doc\explainable survival prediction
model\prism_github\venv_prism\Lib\site-
packages\memory_profiler.py:853(show_results)
    1    0.000    0.000    0.000    0.000 c:\Users\Henry
Pigot\Documents\MEGAsync\2024 post doc\explainable survival prediction
model\prism_github\venv_prism\Lib\site-
packages\memory_profiler.py:775(enable_by_count)
    1    0.000    0.000    0.000    0.000 C:\Users\Henry
Pigot\AppData\Local\Programs\Python\Python311\Lib\inspect.py:391(_has_code_flag)
    1    0.000    0.000    0.000    0.000 C:\Users\Henry
Pigot\AppData\Local\Programs\Python\Python311\Lib\contextlib.py:287(helper)
    1    0.000    0.000    0.000    0.000 C:\Users\Henry
Pigot\AppData\Local\Programs\Python\Python311\Lib\threading.py:1185(is_alive)
    1    0.000    0.000    0.000    0.000 c:\Users\Henry
Pigot\Documents\MEGAsync\2024 post doc\explainable survival prediction
model\prism_github\venv_prism\Lib\site-packages\memory_profiler.py:842(enable)
    2    0.000    0.000    0.000    0.000 c:\Users\Henry
Pigot\Documents\MEGAsync\2024 post doc\explainable survival prediction
model\prism_github\venv_prism\Lib\site-
packages\ipykernel\iostream.py:505(parent_header)
    2    0.000    0.000    0.000    0.000 c:\Users\Henry
Pigot\Documents\MEGAsync\2024 post doc\explainable survival prediction
model\prism_github\venv_prism\Lib\site-
packages\ipykernel\iostream.py:550(_is_master_process)
    1    0.000    0.000    0.000    0.000 c:\Users\Henry
Pigot\Documents\MEGAsync\2024 post doc\explainable survival prediction
model\prism_github\venv_prism\Lib\site-packages\memory_profiler.py:849(disable)
    6    0.000    0.000    0.000    0.000 c:\Users\Henry
Pigot\Documents\MEGAsync\2024 post doc\explainable survival prediction
model\prism_github\venv_prism\Lib\site-
packages\memory_profiler.py:1215(<genexpr>)
    1    0.000    0.000    0.000    0.000 C:\Users\Henry
Pigot\AppData\Local\Programs\Python\Python311\Lib\contextlib.py:104(__init__)
    1    0.000    0.000    0.000    0.000 C:\Users\Henry Pigot\AppData\Local
\Programs\Python\Python311\Lib\threading.py:1118(_wait_for_tstate_lock)
    1    0.000    0.000    0.000    0.000 c:\Users\Henry
Pigot\Documents\MEGAsync\2024 post doc\explainable survival prediction
model\prism_github\venv_prism\Lib\site-packages\memory_profiler.py:689(items)
    1    0.000    0.000    0.000    0.000 C:\Users\Henry
Pigot\AppData\Local\Programs\Python\Python311\Lib\inspect.py:300(ismethod)

```



```

1      0.000      0.000      0.000      0.000 c:\Users\Henry
Pigot\Documents\MEGAsync\2024 post doc\explainable survival prediction
model\prism_github\venv_prism\Lib\site-packages\memory_profiler.py:640(__init__)
2      0.000      0.000      0.000      0.000 {built-in method nt.getpid}
2      0.000      0.000      0.000      0.000 {built-in method sys.settrace}
2      0.000      0.000      0.000      0.000 {method 'items' of 'dict' objects}
1      0.000      0.000      0.000      0.000 {method 'acquire' of
'_thread.lock' objects}
1      0.000      0.000      0.000      0.000 C:\Users\Henry Pigot\AppData\Local
\Programs\Python\Python311\Lib\functools.py:421(_unwrap_partial)
1      0.000      0.000      0.000      0.000 c:\Users\Henry
Pigot\Documents\MEGAsync\2024 post doc\explainable survival prediction
model\prism_github\venv_prism\Lib\site-
packages\ipykernel\iostream.py:138(_event_pipe)
3      0.000      0.000      0.000      0.000 {built-in method builtins.getattr}
1      0.000      0.000      0.000      0.000 C:\Users\Henry
Pigot\AppData\Local\Programs\Python\Python311\Lib\inspect.py:378(isfunction)
2      0.000      0.000      0.000      0.000 {method '__exit__' of
'_thread.RLock' objects}
1      0.000      0.000      0.000      0.000 {method 'startswith' of 'str'
objects}
1      0.000      0.000      0.000      0.000 {method 'disable' of
'_lsprof.Profiler' objects}
2      0.000      0.000      0.000      0.000 {method 'write' of '_io.StringIO'
objects}
1      0.000      0.000      0.000      0.000 {method 'endswith' of 'str'
objects}
2      0.000      0.000      0.000      0.000 {built-in method builtins.len}
1      0.000      0.000      0.000      0.000 {method 'update' of 'dict'
objects}
2      0.000      0.000      0.000      0.000 {method 'get' of
'_contextvars.ContextVar' objects}
1      0.000      0.000      0.000      0.000 {method 'pop' of 'list' objects}
1      0.000      0.000      0.000      0.000 c:\Users\Henry
Pigot\Documents\MEGAsync\2024 post doc\explainable survival prediction
model\prism_github\venv_prism\Lib\site-packages\dateutil\tz\tz.py:74(utcoffset)
1      0.000      0.000      0.000      0.000 C:\Users\Henry
Pigot\AppData\Local\Programs\Python\Python311\Lib\threading.py:568(is_set)
1      0.000      0.000      0.000      0.000 {method 'append' of
'collections.deque' objects}
1      0.000      0.000      0.000      0.000 {built-in method sys.gettrace}
1      0.000      0.000      0.000      0.000 {method 'insert' of 'list'
objects}

```

Memory usage: current=6.3MB, peak=6.9MB
CPU usage: 46.6%

Profiling results saved to: C:\Users\Henry Pigot\Documents\MEGAsync\2024 post doc\explainable survival prediction model\prism_github\models\profiling_results\lebesgue_profile_20240807_100128_samples500_cpu.txt

```
[23]: torch.tensor(x_train.iloc[:500].values, dtype=torch.float32).to(device).shape
```

```
[23]: torch.Size([500, 11])
```